

# Attempt Log Group 6

This project had many attempts to reach its current state. Initially, we tried to create an app through StreamLit/ReactFlow however the current python update made this unusable. After many iterations of trying to correct this without knowing it was the python version breaking things we decided on option 2, making the bowtie through presentation format to simulate the interaction we sought. After completing the in-class assignment, the issue with python version was uncovered and from there progress was much better. I (Jake Rice) was the primary person from the group building this interactive app. Using copilot, StreamLit, and ReactFlow resources I was able to build a basic version. This version had no interaction. However, it did allow for nodes to be created, and for the colors to be changed. From here each bit of interaction was added sequentially.

The first interaction was the creation of the side bar, We decided to opt for this instead of hovering showing details as we felt these extra details might not always be as important during presentations, and if the curser was acting as a pointer, having text pop up as one is following the path of risk would get in the way. This side bar allows for the text in the nodes to be kept minimal and makes viewing the diagram a pleasure.

Next was the ability to click on a threat or consequence and hide the respective barriers. Initially it was made on one click, however, this caused issues with the side bar, if one wanted to show details about the node it would automatically hide the barriers connected with it. To solve this problem, we simply implemented a double click option and assigned it to hiding barriers. This was the simplest solution and makes interaction seamless.

Next was adding the ability to fail a barrier and highlight a path downstream of the failure. This was the hardest part of the project, node ids were used to keep track of what nodes were affected, and where the highlighting should stop. I am very happy with the result of this and the work to implement was worth it. Now one can clearly see the path risk is taken, and the barriers in place to mitigate the consequences associated.

Finally, the legend and symbols were added, symbols were used so a presenter could easily discern the different types of barriers at a glance, so far only human and technological barriers are in place so there are human silhouettes and gears to represent these. Additionally, the legend shows what the red highlighting means, and when a barrier is failed. The final addition came at the very end, the addition of dark and light mode. This came from myself being tired of looking at the white screen of the diagram, so I made a feature so the presenter has the preference to use dark or light mode, automatically switching the colors of node edges, lines connecting, and text to be easy to read in either option.

I would like to take this project in a few directions moving forward. First, building out more pathways would allow for better interaction, for the purpose of this project and being the

sole individual working on this, I was not able to implement many diverse paths. Additionally, I would like to offer different shapes for the threats, hazards, consequences, and barriers. I feel this would make discerning the different nodes much faster. Other additions to interaction would be drop down boxes instead of the side bar, this would help to tell the story of a specific pipeline, instead of the side bar only showing details for one node at a time. The details could drop down from the node itself. Other interesting forms of interaction would be the ability to add arrows or nodes live while presenting. This would assist in the presenter's ability to tell the story and help to highlight key features to specific pathways being discussed.

We hope you enjoy this interactive interpretation of a Bowtie risk diagram; I certainly had fun building it and learning some new skills along the way. This has been added to my GitHub page where it has been made public, here is the link to the [GitHub repo](#), and the link to the [live application](#). Thank you very much for this opportunity!